# QR-013 Trim, sand and bond prep

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| **Doc ID** | QR-013 |
| **Title** | Trim, sand and bond prep |
| **Parent** | CS-009 §1; CS-009 §4; CS-009 §5; CS-009 §6; CS-009 §7.1; CS-009 §7.2; CS-009 §7.3; CS-009 §7.4; CS-009 §7.6; CS-009 §8; CS-009 §9; CS-008 §5; CS-008 §5.1; CS-010 §7.4 |
| **Checked against** | CS-009 Rev D; CS-009 Rev D; CS-009 Rev D; CS-009 Rev D; CS-009 Rev D; CS-009 Rev D; CS-009 Rev D; CS-009 Rev D; CS-009 Rev D; CS-009 Rev D; CS-008 Rev E; CS-008 Rev E; CS-010 Rev D |

Derived from its parent standards. If this and a parent disagree, the parent is right.

## 1. When can I trim a part after it comes out of the mold?

Not before the FEB hold for its resin has run: 48 h for IN2 with AT30 SLOW, 24 h for IN2 with AT30 FAST, 24 h for West 105 with 206 or 205, 36 h for West 105 with 209 (CS-008 §5; CS-009 §7.1 step 1). The holds are team values, longer than the datasheets on purpose.

Also called: trim, trimming, when to trim, cure time, cure hold, FEB hold, demould, demold, de-mould, how long to wait, Trim and finish.

- On the run, "Trim and finish" comes after "Cure and demould" ("Cure" on a Foam wrapped run). That earlier step carries the hold, and the app keeps it locked until the hold has run (CS-009 §7.1, CS-008 §5.1).
- The hold counts from the finish time recorded when the layup or infusion was signed (CS-008 §7 step 6).
- A cold shop runs long. Fingernail-test the flange rather than trusting the clock (CS-008 §5.1).
- Gel is not cure. A part that feels hard can still be soft at the flange (CS-008 §4).
- The datasheet figure for each system and why the holds are longer: QR-001.

## 2. How do I trim a part to the trim line?

Tape the cut path over the trim line, cut outside the line with the Dremel and a reinforced cut-off wheel, then sand to the line (CS-009 §7.1). The dust extractor runs and a second trained person is present for every cut of cured carbon.

Also called: trim, trimming, cut, cutting, edge, trim line, scribe line, Dremel, cut-off wheel, cutting carbon.

1. Check the FEB hold has run (CS-009 §7.1 step 1; QR-013 §1).
2. Mark the cut from the scribe-transferred trim line (CS-009 §7.1 step 2).
3. Tape the cut path. The tape controls chips and shows the line under dust.
4. Put on a half-facepiece respirator with OV/P100 cartridges, nitrile gloves, long sleeves and safety glasses with side shields (CS-009 §6; QR-013 §3).
5. Start the dust extractor at the cut.
6. Check a second trained person, holding the Trimming and finishing training, is present (CS-009 §6; QR-013 §4).
7. Cut outside the line. A Dremel does not steer well enough to cut on it (CS-009 §7.1 step 3).
8. Sand to the line (QR-013 §5). The target is ±1 mm of the line, a team value (CS-009 §8).

Use reinforced cut-off wheels only: plain wheels shatter in CF (CS-009 §5).

Cutting flat panels on the waterjet, the ShopSabre or the bandsaw is CS-009 §7.5.

## 3. What PPE do I need to cut or sand carbon?

Required: a half-facepiece respirator with OV/P100 cartridges, nitrile gloves, long sleeves, safety glasses with side shields, and the dust extractor running at the cut (CS-009 §6). This covers cutting, sanding or machining cured CF, glass or G10.

Also called: dust, carbon dust, CF dust, PPE, respirator, mask, dust mask, P100, gloves, safety glasses, extractor, dust extraction.

- CF dust is the team's worst routine exposure (CS-009 §6).
- Gloves are also for the edges: cut CF splinters (CS-009 §6).
- The shop stocks A + P100 cartridges; 3M 60921 is the OV/P100 cartridge on file (CS-009 §6, `03 Datasheets/3M-spec-60921-OV-P100-Cartridge.pdf`).
- Solvent wipes (IPA, acetone) need nitrile gloves and safety glasses. Acetone gets through Ansell's 0.45 mm latex/neoprene/nitrile 93-800 in 17 min (ASTM F739; 21 min by EN ISO 374), and Ansell gives no acetone time for thin nitrile; the team's brand may differ, so change gloves as soon as acetone wets them (CS-009 §6, `03 Datasheets/ANSELL-chart-TouchNTuff-93-800.pdf`).
- DP420: nitrile gloves and safety glasses with side shields are required, and a respirator with OV/P100 cartridges when ventilation is not adequate (CS-009 §6; QR-013 §8).
- Every chemical, family by family: QR-008.

## 4. Can I cut carbon on my own? The second-person rule

No. Nobody cuts cured carbon without a second trained person present (CS-009 §6). Trained means the person holds the Trimming and finishing training in the app. The rule was set in SN5, and the Composites Lead may change it.

Also called: supervision, buddy rule, second person, two-person rule, trained, training, cleared to cut, Trimming and finishing training.

- The Composites Lead adds the training in People → Catalog ("Add a training", CS standard CS-009) (CS-009 §6).
- The lead may gate the "Trim and finish" step on it from Reports → Layup techniques → Edit steps ("Training to sign") (CS-009 §6).
- On the ShopSabre and the bandsaw the same dust rules apply: extractor running, respirator on, second trained person present (CS-009 §7.5 step 3).
- The bandsaw needs RFS composites bandsaw training, and the Jacobs Hall OMAX waterjet needs Jacobs waterjet training (CS-009 §5).

## 5. What grit do I sand with, and how far can I go?

Only through the sacrificial ply, the 88 spread-tow outer layer; structural plies are not a sanding allowance (CS-009 §7.2, §4). Grits: 120 for trim and keying, 400 for edges, 800 and up before clear coat (CS-009 §5, team grits).

Also called: sand, sanding, grit, sandpaper, wet and dry, finishing, edges, sand-through, sanding budget, sacrificial ply.

1. Read the run's stack before sanding anything (CS-009 §7.2 step 1).
2. In zones under 3 plies, hand-sand only. Count strokes and check often (CS-009 §7.2 step 2).
3. Sand edges 120 then 400 grit (CS-009 §7.2 step 3).
4. Where cosmetics or sealing matter, seal exposed edge fibers with a thin resin wipe. A raw CF edge wicks moisture and splinters into hands.

## 6. Why no machine sander on thin laminate?

Because that is how the SN5 catch cans died: a machine sander on thin laminate at a taper went through the skin and into the substrate (CS-009 §7.2, §1). In zones under 3 plies, hand-sand only, count strokes and

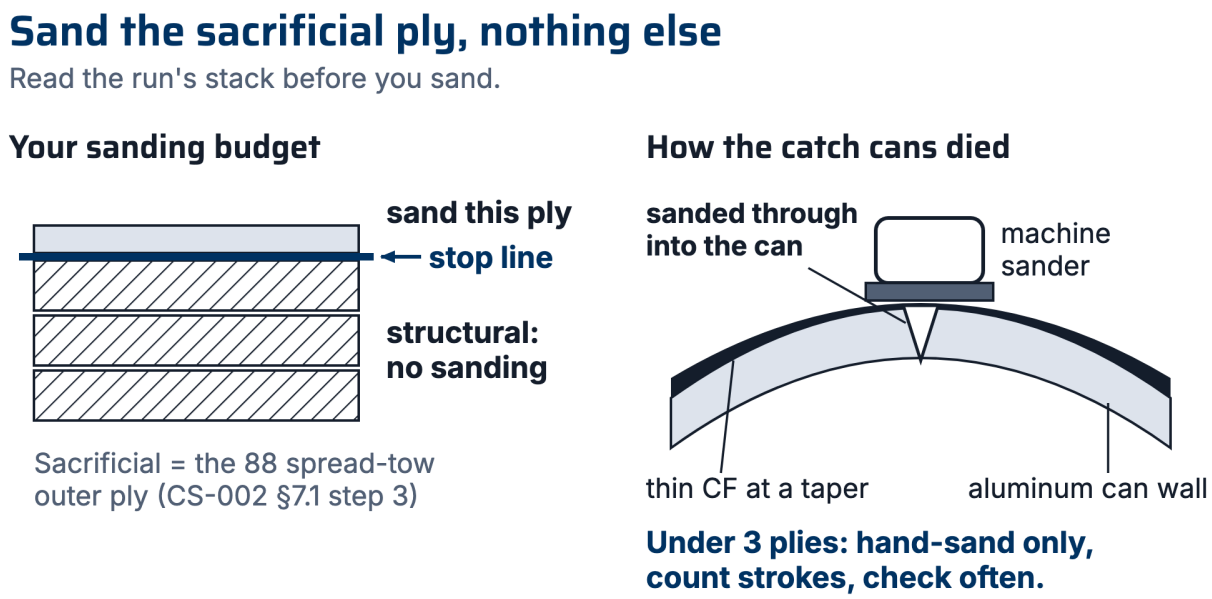



Figure 1: Left: the sacrificial ply is the sanding budget; structural plies are not an allowance. Right: the catch-can failure, a machine sander on thin laminate at a taper, through the skin and into the substrate.

check often.

Also called: orbital sander, DA, random orbit, power sander, machine sanding, thin laminate, catch can, sand-through, sanding through.

- The damage happened on 2026-05-27 (CS-009 §1).
- Thin, tapered or wrapped metal parts are the worst case: sanding punctured the can wall at the taper (CS-010 §7.1).
- Do not sand thin laminates chasing contact (CS-010 §7.4 step 2).
- If weave shows past the plan, stop sanding and raise  on the step (CS-009 §7.2 steps 4 to 7; QR-013 §7).

## 7. I sanded into the weave. What now?

Stop sanding, and report it with  (Report an issue on this step) on "Trim and finish" ("Finish" on a Glass infusion run) (CS-009 §7.2 steps 4 to 7). The run cannot complete until the issue is disposed.

Also called: sand-through, sanded through, sanded too far, weave showing, exposed weave, damage, defect, issue, .

1. Stop sanding.
2. Press  (Report an issue on this step) on "Trim and finish" ("Finish" on a Glass infusion run).
3. Add the spot under "Photos of the defect".
4. Leave the Disposition to the part's engineer or the Composites Lead.

The §8 check is "Sacrificial ply intact / no exposed weave past plan" (CS-009 §8). Weave past the plan is always an issue on the step, not a note.

## 8. DP420 ratio, open time and clamp time

DP420 mixes 2:1 by volume (B:A), 2:0.98 by weight, and the cartridge sets the ratio. Close the joint within the 20 min open time and keep it clamped until handling strength, 2.5 h at 23 °C (73 °F) (CS-009 §7.3, `03 Datasheets/3m-scotch-weld-epoxy-adhesive-dp420-black.pdf`).

Also called: DP420, DP 420, 3M glue, structural adhesive, epoxy glue, two-part glue, work life, open time, cure time, clamp time, handling strength, how long to clamp.

All numbers are datasheet figures from the DP420 TDS (CS-009 §5):

| What | Figure |
|---|---|
| Ratio | 2:1 by volume (B:A), 2:0.98 by weight |
| Open time | 20 min |
| Handling strength (unclamp and move, not load) | 2.5 h at 23 °C (73 °F) |
| Bond strength reported | after 7 days at 23 °C |
| Storage | 16-27 °C (60-80 °F) and 40-60% RH |
| Shelf life | 24 months from manufacture |

- The TDS gives no "24 h full cure". Older notes that say so are wrong (CS-009 Rev D history).
- Load is a separate question with its own team hold (CS-009 §7.6 step 8; QR-013 §11).
- PPE: nitrile gloves and safety glasses with side shields are required; a respirator with OV/P100 cartridges when ventilation is not adequate. You should add indirect vented goggles and a face shield, on the purchase list (CS-009 §6).

## 9. How do I prep a surface for bonding?

Gloves on and ignition sources out first, then solvent, abrade, solvent: wipe both faces with IPA, abrade them with 120 grit until the gloss is gone, then wipe again with clean IPA and let them flash off (CS-009

§7.3 steps 1 to 3). The 120 grit is a team value; the TDS asks for clean, fine-grit abrasive.

Also called: bond prep, surface prep, key, keying, scuff, abrade, IPA wipe, acetone wipe, degrease, prep for glue.

1. Put gloves on and put out ignition sources before any solvent is opened (CS-009 §7.3 step 1).
2. Wipe both bond faces with IPA to lift oil and release agent.
3. Abrade both faces with 120 grit until the gloss is gone.
4. Wipe both faces again with clean IPA.
5. Let the faces flash off.
6. Change into clean gloves and keep them on. Do not touch the bond faces from here (CS-009 §7.3 step 4).

- On cured laminate and metal parts away from any mold, acetone may be used instead of IPA, as the TDS's own Acetone/Abrade/Acetone prep does (CS-009 §7.3 step 3).
- Never acetone on or near tooling board (CS-009 §7.3 step 3).
- On a metal insert or rod end, abrade the full bonded length, not a few file strokes (CS-009 §7.6 step 3).
- No silicone-adhesive tape (Kapton and most amber films) anywhere in the shop: trace silicone kills DP420 adhesion (CS-001 §6).

## 10. How do I glue with DP420, step by step?

Prep both faces (IPA wipe, 120 grit until the gloss is gone, IPA again, let flash off), purge the cartridge in the EPX applicator until both sides flow, fit the mixing nozzle, apply, close and fix the joint within 20 min, and clamp for 2.5 h at 23 °C (73 °F) (CS-009 §7.3 steps 1 to 7).

Also called: DP420, glue, gluing, bond, bonding, adhesive, applicator gun, EPX, mixing nozzle, purge, squeeze-out, fillet.

1. Prep both faces: IPA wipe, 120 grit until the gloss is gone, IPA again, let flash off (CS-009 §7.3 steps 1 to 3; QR-013 §9).
2. Load the duo-pak cartridge in the EPX applicator and remove the cap.
3. Dispense a little until both sides flow evenly.
4. Fit the mixing nozzle.
5. Apply the adhesive.
6. Close and fix the joint within the 20 min open time.
7. Clamp so the squeeze-out forms a continuous fillet.
8. Leave the clamps on until handling strength, 2.5 h at 23 °C (73 °F).
9. Write the bond in the "Trim and finish" note ("Finish" on a Glass infusion run) and add a photo with "Add photos to this step" (CS-009 §9).

A continuous fillet with no starved zones is the acceptance look (CS-009 §8). A joint that will carry load also follows CS-009 §7.6 (QR-013 §11).

## 11. Bonding an insert or rod end into a CF tube or part

Every load-carrying bond shall use DP420; never 5-minute epoxy or superglue (CA). Mask the threads, dam the tube, fixture it until 2.5 h at 23 °C (73 °F), and do not load it before the team hold (CS-009 §7.6). A rear wing rod-end bond failed on track in September.

Also called: insert, rod end, threaded insert, bonded insert, hardware, mount, bracket, load-carrying bond, structural bond, CF tube, carbon tube.

1. Prep both faces: IPA wipe, 120 grit until the gloss is gone over the full bonded length of a metal insert or rod end, IPA again, let flash off (CS-009 §7.3 steps 1 to 3; §7.6 step 3; QR-013 §9).
2. Mask the threads.

3. Dam the tube behind the insert, so adhesive cannot reach the threads or run down the tube. DP420 in a rod-end thread once locked the rod at a set length (CS-009 §7.6 step 4).
4. Purge the cartridge until both sides flow, then fit the mixing nozzle.
5. Hold the bondline at the gap the drawing calls for. The TDS's test bondline is 0.13 to 0.20 mm (5 to 8 mil); the team has not set its own gap (CS-009 §7.6 step 6).
6. Fixture or clamp the joint so it cannot move until handling strength, 2.5 h at 23 °C (73 °F).
7. Do not mount, torque or drive on it before the team hold. That hold is still [lead input needed]; the TDS reports bond strength after 7 days at 23 °C (CS-009 §7.6 step 8). The app does not enforce it.
8. Record adhesive and lot, prep done, time bonded and time first loaded (CS-009 §7.6 step 9; QR-013 §14).

## 12. Can I use 5-minute epoxy, superglue or West 404 for a bond?

Not on anything that carries load. Every load-carrying bond uses DP420, and 5-minute epoxy and superglue (CA) are never used on a load-carrying joint (CS-009 §7.6 steps 1 and 2). West 105/206 thickened with 404 is for fillets and gap filling only.

Also called: 5-minute epoxy, five minute epoxy, quick epoxy, superglue, CA, cyanoacrylate, 404, thickened epoxy, filler, West 404.

- In September testing, an undertray insert turned out to have been bonded with 5-minute epoxy (CS-009 §7.6 step 2).
- West 404 is not for load-carrying bonds unless the Composites Lead approves. The 404 TDS rates it for hardware bonding; the team rule is stricter (CS-009 §5).
- A load-carrying bond is any bond that carries load on the car: an insert, a rod end, a mount, a bracket (CS-009 §4).

## 13. How do I clear coat a part? Mix ratio and coats (LiME LiNE UC4:1)

Part B, the activator, stays shut until its SDS is on file (CS-009 §7.4 step 1). Then scuff, wipe with IPA, label, mix 4 parts UC4:1 clear to 1 part activator, and apply 2 medium wet coats 10 to 15 min apart (CS-009 §7.4).

Also called: clear coat, clearcoat, clear, UC4:1, LiME LiNE, Lime Line, activator, gloss finish, GH60, GH70, GH80.

1. Check Part B's SDS, or the hazard text from the can's label, is in `03 Datasheets/` and CS-009 §6 quotes its PPE. Until then, stop here (CS-009 §7.4 step 1).
2. Scuff the part.
3. Wipe it with IPA.
4. Apply the part label, which the clear coat goes over (CS-009 §7.4 step 2; CS-009 §5).
5. Mix 4 parts UC4:1 clear with 1 part activator (CS-009 §7.4 step 3).
6. Use the mix within its pot life, 2 h at 77 °F / 25 °C (CS-009 §7.4 step 4).
7. Apply 2 medium wet coats, 10 to 15 min apart. To recoat after 24 h, scuff first (CS-009 §7.4 step 5).
8. Leave it to cure before handling: dust free at 30 min, sand and buff at 12 to 24 h air dry, both at 77 °F / 25 °C (CS-009 §7.4 step 6).
9. Record the activator, the number of coats and the time in the "Trim and finish" note ("Finish" on a Glass infusion run) (CS-009 §7.4 step 7).

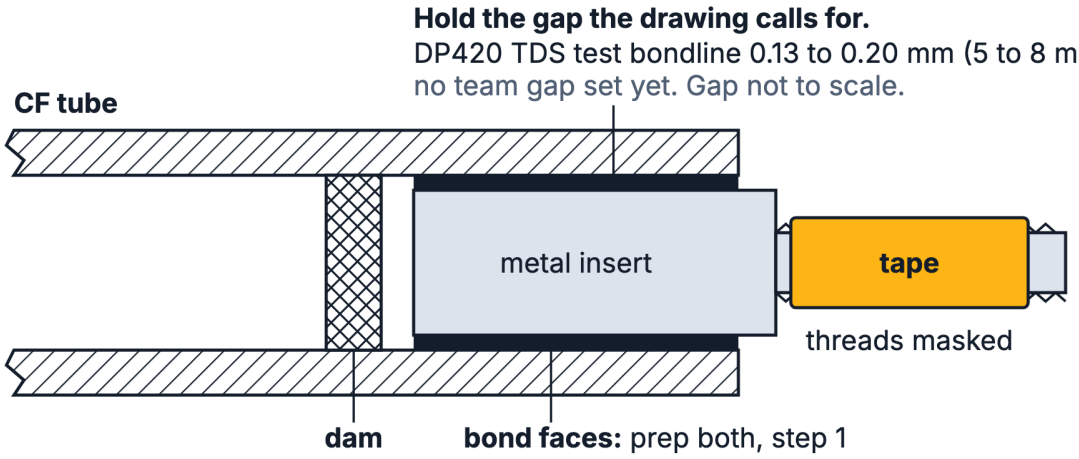
Every number here is a datasheet figure from the LiME LiNE TDS, `03 Datasheets/LIMELINE-TDS-UC4-1-Urethane-Clearco` as copied into CS-009 §7.4. Part A PPE: a respirator with OV/P100 cartridges and nitrile gloves are required; you should wear tightly sealed goggles, which are on the purchase list (CS-009 §6; QR-008 §9).

## 14. Where do I record trimming, sanding and bonding?

In the note on the "Trim and finish" step ("Finish" on a Glass infusion run), then buy off the step, which records your name. Add photos of finished edges and bonds with "Add photos to this step" (CS-009 §9).

## Bonding an insert into CF tube

**Load-carrying bonds: DP420 only.** Never 5-minute epoxy or superglue.




| | | |
|---|---|---|
| 1 | **Prep** | IPA, 120 grit (team value), IPA on both faces.<br>Abrade the full bonded length. Gloves on, hands off. |
| 2 | **Mask, dam** | Tape the threads. Dam the tube behind the insert. |
| 3 | **Bond** | Purge, then fit the nozzle. Close the joint **within 20 min**. |
| 4 | **Clamp** | Clamp or fixture until handling strength, **2.5 h at 23 °C (73 °F)**. |

**No load** — Not on the car, no torque, no driving before the team hold: **[lead input needed]**

**Step note:** adhesive and lot, prep, time bonded, time first loaded.
The app does not enforce the no-load hold. The person bonding does.
Acetone may replace IPA on cured laminate and metal away from any mold (§7.3 step 3).

Figure 2: Section through a CF tube with a bonded insert: solvent, abrade, solvent on both faces; the bondline gap (DP420 TDS test bondline 0.13 to 0.20 mm); a dam inside the tube; masked threads; no load before the team hold.

Also called: record, log, write up, note, step note, photos, buy-off, sign off, Trim and finish, Finish, Execute, first load.

- Measured checks go in the Quality section (Quality checks / acceptance criteria) (CS-009 §9).
- An insert bond records adhesive and lot, prep done, time bonded and time first loaded, in the note on the step where the bond is done: “Trim and finish”, “Finish” on a Glass infusion run, “Execute” on an Other run (CS-009 §7.6 step 9).
- Clear coat records the activator, the number of coats and the time in the same note (CS-009 §7.4 step 7).

| Check | Target | Where it goes |
|---|---|---|
| Trim to scribe | ±1 mm of line (team value) | Quality checks / acceptance criteria |
| No sand-through | Sacrificial ply intact, no exposed weave past plan | Quality checks; weave past plan is a issue |
| Bond squeeze-out | Continuous fillet, no starved zones | Step note and a photo |
| Insert bond record | All four items written; first load after the team hold | Step note |

(CS-009 §8)